

DON'T DIE RETIRED

Strong at 60

The 12-week strength plan for beginners over 50

12-week plan · £14

dontdieretired.com

This guide is general information, not medical advice. Check with your GP or a health professional before starting a new exercise programme, especially if you have a heart condition, high blood pressure, joint problems or have been inactive for a long time. Stop and seek advice if anything hurts.

Strong at 60

A calm, sensible twelve weeks that takes you from never having lifted anything on purpose to training with confidence, twice a week, at home or in a gym.

This plan is for people between roughly 55 and 75 who can walk, get out of a chair and get on with life, and who have decided it is time to start strength training properly. Perhaps for the first time. Perhaps for the first time since a school gym in the 1970s. Either way, you are in the right place, and you are not too late. Muscle responds to training at every age we have ever measured, including in people in their nineties.

You do not need to be fit to start. You do not need to be flexible, thin, sporty or brave. You need two slots of about 40 minutes a week, a little floor space, and a willingness to be a beginner for a few weeks. That is all.

How to use this plan

Read the first four sections once. They explain why this matters, how to stay safe and what the words mean.

Then live in the phase you are on. Each phase is four weeks with two sessions, A and B. Alternate them: A, B, A, B.

Print the progress log and keep it where you train. Ticking a box is more motivating than it sounds.

Every exercise has a home version and a gym version. Pick whichever fits your life this week. You can switch.

Why strength matters after 50

Most of us think of getting older as something that happens to our skin, our hair and our memory. The change that affects daily life most is quieter than any of those. From around the age of 30, and more steeply after 60, adults who do not train tend to lose muscle at roughly **3 to 8 per cent per decade**. Researchers call this sarcopenia. You might call it the reason the shopping feels heavier, the stairs feel steeper and the jar lid will not budge.

Here is the good news, and it is genuinely good. Sarcopenia is not a fixed sentence. A large part of what looks like ageing is actually disuse. When older adults begin a sensible strength programme, they gain muscle and strength at rates that surprise the researchers who measure them. Some of the fastest relative gains recorded in strength studies have come from people in their seventies and eighties who had never trained before. Starting from a low base is, oddly, an advantage: there is a lot of room to improve, and improvement comes quickly.

Four reasons that go beyond looking a bit firmer

- **Bone.** Bone responds to load. When muscles pull hard on bone, the bone slowly remodels to become denser. Strength training is one of the few things shown to slow, and in some cases partly reverse, the loss of bone density that affects both women and men later in life.
- **Grip.** Doctors have noticed that how hard someone can squeeze a handle predicts a surprising amount about their health over the following years. Grip strength is not magic; it is a convenient window onto whole-body strength. Improve the whole body and grip improves with it.
- **Falls.** Most serious injuries later in life start with a fall, and most falls start with a leg that could not react in time or a hip that could not hold. Stronger legs, better balance and a body that has practised getting up from the floor are the best protection we know of.
- **Independence.** Getting out of a low chair, carrying a grandchild, lifting a case into an overhead locker, gardening for an afternoon. None of these need great strength, but all of them need enough, with a margin to spare. Strength training builds that margin.

If you're reading this in the United States

GP = family doctor / primary care physician. The UK Chief Medical Officers' guidance (strength on two days a week, plus balance for over-65s) matches the US Physical Activity Guidelines for Americans, so nothing here changes.

Weights: 2 kg ≈ 4.4 lb, 4 kg ≈ 9 lb, 6 kg ≈ 13 lb, 8 kg ≈ 18 lb, 10 kg ≈ 22 lb, 12 kg ≈ 26 lb, 16 kg ≈ 35 lb, 20 kg ≈ 44 lb. US dumbbell racks run in 5 lb steps; pick the nearest.

Kit prices: a band set is about \$15–20, adjustable dumbbells \$100–300, a 25 lb kettlebell \$30–45. If you're on Medicare Advantage or many supplement plans, SilverSneakers gives free access to thousands of gyms and Y's — the easiest 'gym version' of this plan. Community senior centers often run free or cheap strength classes too.

Protein: 1.2–1.6 g per kg is 0.55–0.75 g per lb of body weight — about 85–115 g a day for a 155 lb person.

What the official guidance says

The UK Chief Medical Officers' physical activity guidelines recommend that all adults, and older adults in particular, do activities that **develop or maintain strength in the major muscle groups on at least two days a week**, alongside regular aerobic activity such as walking. Two days a week is exactly what this

plan asks of you. It is not an ambitious training schedule dreamed up for a book. It is the baseline that the country's most senior doctors think everyone should meet.

“Muscle strengthening activities on at least two days a week, and activities to improve balance, are recommended for all older adults.” —

UK Chief Medical Officers' physical activity guidelines, paraphrased

Fewer than one in three adults over 65 in the UK actually manages this. Finishing this plan puts you in a minority, and it is a very good minority to be in.

Before you start: safety and sense

Strength training, done the way this plan describes, is one of the safest forms of exercise there is. Injury rates are far lower than for running or racket sports. Even so, a few minutes of sensible preparation is worth it.

Do I need to see my GP?

If you are generally well and already active, walking, gardening and getting about without trouble, you can start. If any of the following apply to you, have a conversation with your GP or practice nurse first. It is rarely a reason not to train; it is usually a reason to train with a little extra care.

- ☐ A heart condition, or chest pain or breathlessness on exertion
- ☐ Blood pressure that is not yet controlled by treatment
- ☐ Dizziness, fainting or balance problems that are new or unexplained
- ☐ A recent operation, fracture or joint replacement
- ☐ A diagnosis of osteoporosis (you should still train, but some movements need adapting)
- ☐ Diabetes with complications affecting your feet, eyes or kidneys
- ☐ Any condition where you have been told to avoid straining or lifting
- ☐ Simply not being sure. Asking costs nothing.

Red flags: stop and seek advice

Stop the session and speak to a doctor, or call 111 or 999 if severe, if you experience any of the following:

Chest pain, pressure or tightness, or pain spreading to the arm, jaw or back

Sudden breathlessness out of proportion to the effort

Dizziness, faintness or feeling you might black out

A sudden, severe headache

Sharp or shooting pain in a joint or the spine, as opposed to muscle effort

Numbness, tingling or sudden weakness in a limb

These are uncommon. Knowing them means you never have to worry about them.

Everyday sense

- Train in shoes with a flat, firm sole, or barefoot on a non-slip floor. Cushioned running trainers make balance harder.
- Have something sturdy to hold within reach for balance work: a kitchen counter, a heavy table, a stair rail.
- Breathe. Breathe out during the effort, breathe in as you return. Never hold your breath and strain; it spikes blood pressure.
- Muscle effort feels like a burning, tiring ache in the middle of the muscle. Joint pain feels sharp, pinching or deep and localised. The first is normal. The second means stop and adjust.
- If you feel unwell, skip the session. Nothing is lost. The plan will still be there.

Kit: what you need and what it costs

The plan is designed to work with almost nothing in Phase 1, and with modest, adjustable equipment afterwards. You do not need a home gym. You do not need a smart mirror. Here are the realistic options, with typical UK prices at the time of writing.

Option	What it is	Rough cost	Good for
Household items	A sturdy chair, bottom stair, two shopping bags, filled water bottles, a broomstick	£0	Phase 1, and back-up for every phase
Resistance band set	4 or 5 looped or handled bands of increasing thickness, ideally with a door anchor	£10 to £15	Phase 1 pulling and pressing; travel; warm-ups throughout
Pair of fixed dumbbells	Two matched weights, for example 4 kg or 6 kg each	£20 to £40 per pair	Phase 2 starter kit; you will outgrow them
Adjustable dumbbells	Dial or pin systems from around 2 kg up to 20 kg or more per hand	£80 to £250	Phases 2 and 3 at home; the best long-term home purchase
Kettlebell	One cast-iron bell, 8 kg or 12 kg to start	£20 to £40	Deadlifts, carries, goblet squats; very versatile
Gym membership	Council leisure centre or budget chain	£20 to £40 per month	All phases; machines make Phase 2 simple; staff on hand

Our honest recommendation

Buy the band set now, because it costs less than a takeaway and you will use it for years. Decide at the end of Phase 1 whether you want to train at home or join a gym. If home, a pair of adjustable dumbbells or a single 12 kg kettlebell will carry you through the rest of the plan and well beyond. If gym, most council leisure centres offer a free induction; take it, and tell them you are following a beginner's plan and want to learn the leg press, chest press, seated row, lat pulldown and where the dumbbells live.

How to warm up

A warm-up is not a ritual. It raises your body temperature, gets fluid moving in the joints and rehearses the movements you are about to load. Five to eight minutes is plenty. Do the same warm-up before every session so it becomes automatic.

1. **Two to three minutes of easy movement.** March on the spot, walk briskly round the house, climb the stairs twice, or pedal a gym bike gently. You want to feel warmer and breathe a little deeper, not puff.
2. **Ten slow arm circles** each way, then **ten shoulder shrugs and rolls.** Loosen the neck by turning gently to each side.
3. **Ten hip circles** each way with hands on hips, as if stirring a large pot.
4. **Ten bodyweight sit-to-stands** from a chair, slowly, no hands if you can.
5. **Ten hip hinges** with a broomstick or your hands sliding down your thighs: push the hips back, keep the back long, return.
6. **Ten band pull-aparts** or, without a band, ten slow chest-openers with arms wide.
7. **A rehearsal set.** Do the first exercise of the session with no weight, or the lightest weight, for eight easy reps. Now you are ready.

A note on stretching

Long static stretches before training do not prevent injury and can slightly reduce strength. Save them for after, or for a separate relaxing session in the evening.

If something feels stiff before you start, more gentle movement is the answer, not holding a stretch.

The language of training, in plain English

Strength training has its own vocabulary, and much of it exists to make simple ideas sound impressive. Here is everything you need.

Reps and sets

A **rep** (repetition) is one complete movement: one squat down and up, one press out and back. A **set** is a group of reps done one after another before you rest. When the plan says 3 × 10, it means three sets of ten reps, with a rest between each set. When it says 3 × 8 to 12, it means aim for somewhere between eight and twelve reps in each set.

How hard should it feel? Reps in reserve

This is the most important idea in the whole plan, so it gets its own space. You will never be asked to lift to failure, to grind out a rep with your face turning purple, or to guess percentages of a maximum you have never tested. Instead we use **reps in reserve**, often shortened to RIR.

At the end of each set, ask yourself: how many more reps could I have done with good form before I had to stop? That number is your reps in reserve. For most of this plan you should **stop when you have two or three reps left in you**. The set should feel like real work, your muscles should be tiring, but you should finish it knowing you could have done a couple more. That is the sweet spot where you get stronger without unnecessary strain or excessive soreness.

Reps in reserve	How it feels	When to use it
4 or more	Easy. You could keep going for a while.	Warm-ups, first week of a new phase, coming back after illness
3	Working, but comfortable. Form is perfect.	Phase 1 and most of Phase 2
2	Hard. The last rep slows down noticeably.	Phase 3 and the last week of each phase
1	Very hard. Next rep might not happen.	Not needed in this plan
0	Failure. You tried and could not.	Avoid. No benefit for beginners, more soreness and risk

Rest

Rest between sets for **60 to 90 seconds** in Phases 1 and 2 and **90 seconds to two minutes** in Phase 3. Rest is not wasted time; it is where your muscles recover enough to do the next set properly. Sit down if you like. Between exercises take as long as you need to set up. Between sessions, leave at least one full day: Monday and Thursday, or Tuesday and Friday, works well.

Progression: the rules that make it work

Doing the same thing every week maintains what you have. To build, you have to gradually ask a little more. This is called progressive overload, and it is why the plan is twelve weeks rather than one week repeated twelve times. Follow these rules and progression takes care of itself.

1. **Add a rep before you add weight.** If the plan says 8 to 12 reps, start at 8 with a weight that leaves 2 or 3 reps in reserve. Each session, try to add a rep to each set. When you can do 12 reps on all sets with 2 reps still in reserve, increase the weight and drop back to 8.

2. **Increase weight in the smallest step you have.** One band thicker, 1 to 2 kg per dumbbell, one plate on a machine. Small steps add up to large totals over twelve weeks.
3. **Sets before intensity.** In Phase 1 you go from two sets to three sets before anything else changes. More total work, same difficulty per set.
4. **Slower counts as harder.** If you cannot add weight, take three seconds to lower the weight instead of one. Controlled lowering builds strength and is kind to joints.
5. **Never progress two things at once.** More weight, or more reps, or more sets. Not all three in the same week.
6. **Write it down.** Progression is impossible to manage from memory. The log at the back of this plan is not optional.

The six patterns and fourteen exercises

Every useful strength exercise is a version of one of six movement patterns. Learn the patterns and you understand every gym in the world, every piece of kit and every workout you will ever be shown. The plan builds all six.

- **Squat.** Sitting down and standing up. Knees and hips bend together.
- **Hinge.** Bending at the hips with a long back to pick something up. Hips move, knees mostly do not.
- **Push.** Moving something away from you: forwards (a door) or overhead (a case into a locker).
- **Pull.** Moving something towards you: rowing a heavy drawer open, or pulling yourself up a rail.
- **Carry.** Holding a load and walking. Shopping, luggage, a toddler.
- **Single-leg and balance.** Stairs, stepping over things, standing on one foot to put a sock on.

Below are the fourteen exercises used across the three phases, each with a home and gym version, a single cue to think about, and the mistake we see most often. Read the row before the first time you do an exercise, then again after a week when it will make more sense.

Exercise	Home version	Gym version	Cue	Common mistake
Squat	Chair squat: sit back to touch the chair and stand	Goblet squat (weight held at chest) or leg press	Sit back and down as if finding a chair; knees follow toes	Heels lift or knees drift inwards; going too fast
Hinge (deadlift)	Broomstick hip hinge, or band good morning	Dumbbell or kettlebell Romanian deadlift	Push hips back, shins still, back long as a table	Rounding the back; bending knees so it becomes a squat
Glute bridge	Floor bridge, feet flat	Weighted bridge or hip thrust off a bench	Push through heels, squeeze at top, ribs down	Arching the lower back instead of lifting hips
Horizontal push	Wall press-up, then counter, then incline on a low step	Dumbbell floor or bench press; chest press machine	Elbows about 45 degrees, body one straight line	Hips sagging; elbows flared straight out
Vertical push	Band overhead press (stand on band)	Seated or standing dumbbell shoulder press	Ribs down, press up and slightly back	Arching the lower back; shrugging into the ears
Horizontal pull (row)	Band row with door anchor, or towel row round a post	One-arm dumbbell row; seated cable row	Lead with the elbows, squeeze shoulder blades together	Shrugging; using the lower back to swing
Vertical pull	Band lat pulldown, band anchored high	Lat pulldown machine; assisted pull-up	Pull elbows down to the ribs, chest tall	Leaning right back; pulling with the arms only
Band pull-apart	Band held at shoulder height, pull to chest	Reverse fly machine or light dumbbell reverse fly	Chest up, pull band to collarbones, slow return	Bending elbows a lot; letting shoulders round
Carry	Farmer's carry with two shopping bags or bottles	Farmer's carry with dumbbells or kettlebells	Stand tall, shoulders down, walk slowly and quietly	Leaning to one side; rushing; gripping too loosely
Step-up	Bottom stair, hand on rail as needed	Box or bench step-up, dumbbells in hands	Whole foot on the step, drive through the heel	Pushing off hard with the back foot; knee caving
Split squat	Feet staggered, one	Same, holding	Drop straight down,	Leaning forwards;

Exercise	Home version	Gym version	Cue	Common mistake
	hand on chair	dumbbells	front heel heavy	back knee shooting outwards
Single-leg balance	Stand on one leg by the counter	Same, or on a soft mat for a challenge	Soft knee, eyes on one fixed point, breathe	Locking the knee; holding breath; gripping the counter
Calf raise	Bottom stair, heels off the edge, hand on rail	Calf raise machine or holding dumbbells	Slow up, pause, slower down	Bouncing; not going through full range
Dead bug	Lying on back, opposite arm and leg lower slowly	Same, or Pallof press with a cable	Lower back pressed into floor, breathe out as you lower	Rushing; lower back lifting off the floor

Adapting for osteoporosis or a stiff spine

If you have low bone density, or your GP has advised you to avoid spinal flexion, replace the dead bug with a bird dog (on hands and knees, extend opposite arm and leg) and keep all hinges to a very comfortable range.

Avoid any movement that rounds the spine under load. Everything else in this plan is not only safe for bone but positively good for it.

Phase 1: Foundations (weeks 1 to 4)

The aim of Phase 1 is not to get tired. It is to learn the movements so well that you could do them half asleep, and to build the habit of turning up twice a week. Use bodyweight, bands and household items. If you are joining a gym, you can still do Phase 1 there with bands and a mat, or in your living room. Rest 60 to 90 seconds between sets.

Session A	Sets × reps	Notes
Chair squat	2 to 3 × 8 to 12	Touch the chair lightly, do not sit fully
Incline press-up	2 to 3 × 8 to 12	Wall in week 1, counter by week 3 if easy
Band row	2 to 3 × 10 to 15	Door anchor at waist height
Glute bridge	2 to 3 × 10 to 15	Two-second squeeze at the top
Farmer's carry	2 to 3 × 20 to 30 metres	Shopping bags with tins; equal weight each side
Single-leg balance	2 × 20 to 30 seconds each side	Fingertips on counter to begin

Session B	Sets × reps	Notes
Broomstick hip hinge	2 to 3 × 10 to 12	Stick along the spine, touching head, back and tailbone
Band overhead press	2 to 3 × 10 to 12	Stand on the band, handles at shoulders
Band lat pulldown	2 to 3 × 10 to 15	Anchor above a door; kneel or sit
Step-up	2 to 3 × 8 to 10 each leg	Bottom stair; alternate the leading leg each set
Calf raise	2 × 12 to 15	Slow; pause at the top
Dead bug	2 × 6 to 8 each side	Breathe out as arm and leg lower

Week by week

Week	Sets	Effort	Focus
1	2	4 reps in reserve	Learn the movements. Film yourself if you can.
2	2	3 to 4 reps in reserve	Slightly harder band or slightly lower press-up surface
3	3	3 reps in reserve	Third set added. Expect it to feel like real work.
4	3	2 to 3 reps in reserve	Add reps. If every set hits the top of the range, you are ready for Phase 2.

End of Phase 1 check

Can you do 12 chair squats without using your hands? Can you hold a single-leg balance for 30 seconds on each side? Are the band exercises now easy on your thickest band?

If yes to all three, move on. If not, repeat week 4. There is no prize for rushing, and nobody is watching.

Phase 2: Adding load (weeks 5 to 8)

Now you introduce external weight. At home that means dumbbells or a kettlebell; in a gym it means the machines and the dumbbell rack. The movements are the same ones you have practised for four weeks, so the only new thing is the weight in your hands. Start lighter than you think you should; the first week of any phase uses 3 to 4 reps in reserve. Rest 60 to 90 seconds between sets, a little longer after the deadlift.

Session A	Home	Gym	Sets × reps
Squat	Goblet squat with dumbbell or kettlebell	Leg press or goblet squat	3 × 8 to 12
Horizontal push	Dumbbell floor press	Chest press machine or dumbbell bench press	3 × 8 to 12
Horizontal pull	One-arm dumbbell row, hand on chair	Seated cable row	3 × 8 to 12
Hinge	Kettlebell or two-dumbbell deadlift from a low step	Same, or trap bar if available	3 × 8 to 10
Carry	Farmer's carry with dumbbells	Farmer's carry with dumbbells	3 × 20 to 30 metres
Single-leg	Split squat, bodyweight or light dumbbells	Same	2 × 8 each leg

Session B	Home	Gym	Sets × reps
Hinge	Dumbbell Romanian deadlift	Same	3 × 8 to 10
Vertical push	Seated dumbbell shoulder press	Same, or shoulder press machine	3 × 8 to 12
Vertical pull	Band lat pulldown, thickest band	Lat pulldown machine	3 × 10 to 12
Single-leg	Step-up holding dumbbells	Box step-up with dumbbells	3 × 8 each leg
Pull-apart	Band pull-apart	Reverse fly machine	2 × 12 to 15
Calf raise	Stair, holding one dumbbell	Calf raise machine	2 × 12 to 15
Core	Dead bug	Dead bug or Pallof press	2 × 8 each side

Week by week

Week	Effort	What changes
5	3 to 4 reps in reserve	Find your starting weights. Err light. Write every weight down.
6	3 reps in reserve	Add a rep to each set where you can
7	2 to 3 reps in reserve	Where you hit the top of the rep range, increase the weight one small step and drop to the bottom
8	2 to 3 reps in reserve	Consolidate. Aim to beat week 7 by one rep or one small weight step per exercise

A typical starting point for a healthy beginner in this age group is 6 to 10 kg for the goblet squat, 4 to 8 kg per hand for presses and rows, 8 to 16 kg total for the deadlift and 6 to 10 kg per hand for carries. These are only reference points. Your body, not the number, decides. If 4 kg leaves you with 3 reps in reserve at 10 reps, 4 kg is the right weight.

Phase 3: Building strength (weeks 9 to 12)

In Phase 3 every session includes all six patterns, the rep ranges come down a little so the weights can go up, and the rests get longer. This is where you start to feel like someone who trains, rather than someone who is trying exercise. Rest 90 seconds to two minutes between sets of the big movements (squat, hinge, presses, rows) and 60 seconds for the rest.

Session A	Home	Gym	Sets × reps
Squat	Goblet squat, heavier	Goblet squat or leg press	3 × 6 to 10
Hinge	Dumbbell Romanian deadlift	Same	3 × 6 to 10
Horizontal push	Dumbbell floor press	Dumbbell bench press	3 × 6 to 10
Horizontal pull	One-arm dumbbell row	Seated cable row or dumbbell row	3 × 8 to 10
Carry	Suitcase carry, one hand only	Same	3 × 20 metres each side
Single-leg	Split squat with dumbbells	Same	3 × 8 each leg

Session B	Home	Gym	Sets × reps
Single-leg	Step-up with dumbbells, higher step	Box step-up, dumbbells	3 × 8 each leg
Hinge	Kettlebell deadlift, heavier	Same, or trap bar deadlift	3 × 6 to 8
Vertical push	Standing dumbbell shoulder press	Same	3 × 6 to 10
Vertical pull	Band lat pulldown with pause	Lat pulldown machine	3 × 8 to 10
Carry	Farmer's carry, heavier	Same	3 × 30 metres
Balance	Single-leg stand with slow head turns	Same, or on a soft mat	3 × 30 seconds each side
Core	Dead bug, slow	Pallof press	2 × 8 each side

Week by week

Week	Effort	What changes
9	3 reps in reserve	New rep ranges mean slightly heavier weights. Find them carefully.
10	2 to 3 reps in reserve	Add reps. Slow the lowering phase to three seconds on squats and presses.
11	2 reps in reserve	Increase weight wherever you hit the top of the range. This is your hardest week.
12	3 reps in reserve	Ease off slightly and enjoy it. Then compare your log to week 1.

Why week 12 is easier than week 11

Strength is built during recovery, not during the session. A slightly lighter final week lets everything you have done over eleven weeks settle in, and you finish feeling strong rather than flattened.

It also means you end the plan wanting more, which is exactly the right way to end it.

Your 12-week progress log

Print this page, or copy it into a notebook. After each session, note the date, the heaviest or hardest thing you did, and how you felt. Over twelve weeks this page becomes the most convincing evidence you will ever see that you are getting stronger.

Week	Phase	Session A date	Session B date	Hardest thing this week	Energy 1 to 5	Notes
1	1					
2	1					
3	1					
4	1					
5	2					
6	2					
7	2					
8	2					
9	3					
10	3					
11	3					
12	3					

Three simple markers to test in week 1 and week 12

- ☐ **Sit-to-stand in 30 seconds.** From a standard dining chair, arms crossed, how many times can you stand fully and sit back down in 30 seconds? Write the number down.
- ☐ **Single-leg stand.** How many seconds can you stand on each leg, no support, before you need to touch down? Stop at 60.
- ☐ **Carry test.** Fill two bags to a weight you can carry for 20 metres comfortably. In week 12, try the same bags plus an extra tin in each.

Recovery, protein and sleep

You have probably heard that muscle is built in the kitchen or in bed rather than in the gym. Like most slogans it is half true. The training is the signal; the food and rest are what your body uses to answer it. Neither works without the other, but you do not need to become obsessive about any of it.

Protein

As we age, muscle becomes less responsive to protein, a phenomenon researchers call anabolic resistance. The practical result is that older adults who train seem to do better with a bit more protein than the general population guideline. A range of roughly **1.2 to 1.6 grams of protein per kilogram of body weight per day** is commonly recommended for older adults who do strength training. For a 70 kg person that is about 85 to 110 grams a day. This is not a prescription, and if you have kidney disease or have been given specific dietary advice, follow that instead.

In practice it means having a proper source of protein at each of three meals rather than one large portion at dinner: eggs or Greek yoghurt at breakfast, beans, fish, cheese or chicken at lunch, and a palm-sized portion of meat, fish, tofu or lentils at dinner. A glass of milk or a yoghurt after training is a pleasant, cheap habit that many people find helps.

Food	Typical portion	Protein, roughly
Eggs	2 large	13 g
Greek yoghurt	150 g pot	15 g
Chicken breast	120 g cooked	35 g
Tinned tuna or salmon	1 small tin	20 to 25 g
Lentils or chickpeas	200 g cooked	15 to 18 g
Cottage cheese	150 g	17 g
Tofu	150 g	18 g
Semi-skimmed milk	300 ml glass	10 g

Sleep

Sleep is when growth hormone is released and when the tiny repairs to muscle fibres that make them stronger actually happen. Most adults do best on seven to nine hours, and older adults often find sleep lighter and more fragmented. Strength training itself tends to improve sleep quality within a few weeks, so this often takes care of itself. Avoid training within two hours of bed if you find it leaves you wired; a morning or early afternoon session suits most people.

Rest days

On the five days you are not lifting, keep moving. Walk, garden, swim, cycle, dance. Gentle activity speeds recovery; total rest slows it. What you should not do is add a third or fourth heavy session because you are enjoying it. Two sessions is the plan for a reason. If you want more, add a walk.

Troubleshooting

Muscle soreness a day or two later

Delayed onset muscle soreness, or DOMS, is the dull ache that arrives 24 to 48 hours after new or harder exercise. It is normal, harmless and unrelated to how much progress you are making. It is worst after week 1, week 5 and week 9, when something new starts, and fades within a fortnight of each. Gentle movement, a warm bath and time are the cures. Training again while mildly sore is fine and usually helps. If you are so sore you struggle to sit down, you did too much: drop to 4 reps in reserve next session.

Joint pain

Sharp, pinching or lingering pain in a joint is different from muscle effort and should not be pushed through. First, check your form against the cue in the exercise table. Second, reduce the range of motion: half a squat with no knee pain is worth more than a full one with it. Third, swap the variation: a leg press instead of a squat, a floor press instead of a bench press, a band row instead of a dumbbell row. If a joint hurts for more than a week, or swells, see a physiotherapist. Most joint issues in beginners are irritation from doing slightly too much slightly too soon, and settle with a small adjustment.

Plateau

Around week 7 or 10, progress often stalls for a week. This is normal; the quick early gains come from your nervous system learning to use the muscle you already have, and the slower gains of actual new muscle take longer to appear. Do not add weight in frustration. Instead: check you are actually eating enough protein, check you are sleeping, slow the lowering phase of each rep to three seconds, and give it two more weeks. Plateaus almost always break on their own.

Missed a week

Life happens. One missed week changes nothing; pick up where you left off, perhaps at 3 to 4 reps in reserve for the first session back. Two missed weeks: repeat the last week you completed, then continue. Three or more weeks: go back to the start of the phase you were in. Never try to catch up by doing extra sessions. The plan is not a race, and you cannot fall behind because there is no one to fall behind.

Holidays

Take the band set. It fits in a side pocket. Do Session A and B from Phase 1 in the hotel room, or do one full-body band session and lots of walking. Two weeks of maintenance costs you nothing; two weeks of guilt costs you motivation. Enjoy the holiday.

Illness

A cold above the neck: train if you feel like it, ease off if you do not. Anything with a fever, chest symptoms or that leaves you drained: rest until you have been well for two days, then return at 4 reps in reserve. Some antibiotics can weaken tendons temporarily; if you are prescribed some, ask the pharmacist whether you should hold off on training for a while.

Feeling wobbly or dizzy

Standing up quickly after a set on the floor can drop blood pressure briefly. Get up in stages. Make sure you have eaten something in the previous two to three hours and are drinking water. If dizziness happens regularly or comes with anything on the red flag list, speak to your GP.

A note for women

Strength training matters for everyone, but the case for women after 50 is particularly strong, and it is one that has been underplayed for decades. Through the menopause and in the years after it, the fall in oestrogen accelerates the loss of both muscle and bone. Women can lose a substantial share of their bone density in the years around menopause, and low bone density is common by 60. Falls, fractures and the loss of independence that follows are not an inevitable part of being a woman in later life, but they are common enough that prevention deserves real attention.

Strength training addresses this more directly than any other single lifestyle change. Loading bone through the hips and spine, as squats, deadlifts, step-ups and carries do, is precisely the stimulus bone needs to maintain and rebuild density. The muscle gained gives the strength and balance that stop a stumble becoming a fall in the first place. Many women in this age group also find that training improves sleep, mood and body confidence within a few weeks, though individual experience varies.

- You will not get bulky. Building large muscles requires years of dedicated effort, a great deal of food and, usually, a young man's hormones. What you will get is firmer, denser and stronger.
- Lift heavier than you think a woman your age should. Very light weights for many reps do little for bone. The 6 to 12 rep ranges in this plan, with weights that leave 2 to 3 reps in reserve, are what bone responds to.
- If you have been diagnosed with osteoporosis, this plan is still for you, with the adaptations in the exercise section. Discuss it with your GP or a physiotherapist, and consider asking for a referral to a strength and balance class alongside it.
- If you are on HRT, nothing here changes. If you are considering it, strength training is complementary, not a substitute.
- Pelvic floor: if you notice leaking during heavier lifts, breathe out as you lift rather than holding your breath, reduce the load slightly, and speak to your GP about a pelvic health physiotherapy referral. It is very treatable.

A note for men

Men tend to start later life with more muscle and denser bone than women, which brings a hidden risk: it makes the decline easier to ignore. Testosterone falls slowly from the forties onwards, muscle follows, and many men do not notice until the moment they cannot do something they have always done. Men are also less likely to be checked for osteoporosis, despite a meaningful number of men over 50 breaking a bone because of it.

- You may remember being much stronger. Your ego may want to start there. Do not. Start where this plan says, and let your log, not your memory, decide when to add weight.
- Pride is the most common cause of injury we see in men returning to training. A set left with 3 reps in reserve is a set that lets you come back on Thursday.
- Grip, carries and deadlifts tend to be where men progress fastest and enjoy themselves most. Lean into that.
- If you have prostate treatment or take medication that affects hormones, strength training is especially valuable for preserving muscle and bone. Ask your consultant; almost all will encourage it.
- Belly fat carried around the middle responds well to strength training combined with modest dietary changes. It is not vanity; it is the type of fat most linked with heart disease and diabetes.

What next

You have finished twelve weeks. Look back at your week 1 log and your week 12 log. The numbers will tell you a story that no one can argue with: more reps, more weight, longer holds, heavier bags. You are, measurably, a stronger person than you were three months ago. Most people who reach this point tell us the surprising part was not the strength but the confidence: on stairs, in crowds, on uneven paths, carrying things, getting up off the floor without a second thought.

Keep going

The simplest next step is to repeat Phase 3 for another twelve weeks, adding weight when you hit the top of each rep range, exactly as you have learned. Two sessions a week, for the rest of your life, is a completely reasonable ambition. If you want variety, rotate the exercise variations in the table: swap goblet squats for leg press for a block, floor press for bench press, dumbbell rows for cable rows. The patterns stay the same. The patterns are the point.

- ☐ Repeat your three markers: sit-to-stand in 30 seconds, single-leg stand, carry test. Write them next to week 1.
- ☐ Buy the next size of dumbbell or kettlebell, or ask at the gym about moving to the barbell. You have earned it.
- ☐ Put your two sessions in the diary for the next twelve weeks now, while you are feeling pleased with yourself.
- ☐ Tell someone. Partners, friends and grown-up children over 50 need to hear this is possible from a person they know.

Stay with us

Strong at 60 is one part of what we do at dontdieretired.com. The free weekly newsletter carries a short, practical piece every week on strength, walking, balance, sleep, food and keeping your mind sharp, written for people exactly like you, without hype and without selling you supplements. Sign up on the site if you have not already.

If you would like company, the members' circle is a small, friendly group of readers who share their logs, ask questions and keep each other turning up on Tuesdays and Fridays. There is a monthly live question-and-answer session, a library of follow-on plans, and a good deal of gentle encouragement. Details are on the site under Members.

One last thing

The best predictor of how strong you will be at 75 is not how strong you are at 60. It is whether you are still training.

Two sessions a week. Add a rep before you add weight. Stop with a couple in reserve. Write it down.

That is the whole secret, and now it is yours.

This plan is general information for healthy adults and is not a substitute for individual medical advice. If you have a medical condition, are recovering from injury or surgery, or are unsure whether this plan is suitable for you, please speak to your GP or a qualified health professional before starting.